

GRAINGUARD · REQUIREMENTS

GrainGuard requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 decisions (GGD-DDR-001); R13 redefined per bin with the receiver per farm; R4 plenum EMC basis; reference case corrected; status from GGD-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

GrainGuard requirements

These requirements were checked by calculation at TRL 3 in GGD-CAL-001. Targets are not yet validated with farmers and will be revised after co-design visits (see GGD-PRB-001). One requirement is not met (R8, battery recovery), four are at risk (R2, R4, R6, R9) and two cannot be verified at TRL 3 (R3, R10); Table 2 gives the status of each. Version 0.3 records the decisions in GGD-DDR-001: R13 is redefined per bin with the farmhouse receiver costed per farm (D1), and R4 states how the plenum-air EMC is found (D3, D9).

The **reference case** is a 5.49 m (18 ft) diameter corrugated steel bin with a 5.6 m eave above the pad, a full perforated floor 0.4 m above the pad and grain level with the eave (5.2 m deep): about 88.8 t (3,493 bu) of shelled corn stored at a target of 15.0 % moisture, wet basis, in the US Midwest (GGD-DDR-001, D11), with one existing aeration fan giving about 0.330 m³/s (699 cfm, 0.2 cfm/bu).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Grain temperature profile	6 or more levels on one center cable, 1.0 m or less apart; ± 0.3 °C; one reading every 10 min	Sensor datasheet; geometry check in the model
R2	Grain moisture estimate from interstitial air	Within ± 0.8 percentage points, wet basis, for interstitial RH of 20 % to 75 % and grain at 0 °C to 30 °C, from readings taken 6 h or more after the fan stops	EMC error analysis with sensor tolerance; later comparison with oven or meter samples
R3	Ambient air measurement	± 0.3 °C and ± 2 % RH inside a radiation shield; sampled every 5 min; decisions on 30 min averages	Datasheet and shield error estimate
R4	Fan decision rule	Automatic mode runs the fan only when the plenum-air EMC (from the ambient humidity ratio and the plenum temperature, which is ambient plus the fan heat: measured when the plenum probe option is fitted, otherwise predicted) is at or below the target moisture (hold mode), or when plenum air is 5 °C or more below the mean grain temperature and its EMC is no more than 1.0 point above target (cool mode); minimum run 30 min, minimum off 15 min, 4 starts per hour or fewer	Logic review against extension guidance; later simulation over a season of weather data
R5	Fail-safe control	In automatic mode the fan stops within 60 s of controller power loss or when sensor data are more than 30 min old; the hand position of the hand-off-auto switch runs the fan with GrainGuard removed; fan running is confirmed by current within 30 s and a mismatch raises an alarm	Circuit review; later bench test
R6	Heating alert	Alarm at the farmhouse within 15 min when any pod rises 3 °C or more above its 7-day minimum with the fan off, or sits 5 °C or more above the bin median	Logic review; later heated-core test
R7	Radio link	Bin to farmhouse over 1 km or more with one farm building in the path; 95 % or more of packets delivered	Link budget; later site survey
R8	Power autonomy	5 days or more with no solar input at -20 °C with the fan running continuously; recovers from 50 % charge to full in 3 winter days of 1.5 peak sun hours	Power budget
R9	Environment	Operate at -30 °C to +50 °C; controller IP66; pods dust-tight (IP6X) with a vapor-permeable membrane; survive one phosphine fumigation per season	Datasheets and design review; later exposure test
R10	Mechanical strength	Rope minimum breaking load 4 times or more the design pull-down force of 2.5 kN (estimate); roof hanger rated by the bin maker for the design force	Force estimate; bin maker's data
R11	Install and service without grain entry	Installed with the bin empty, from the roof manhole and floor under a confined-space and fall-protection procedure; no GrainGuard task needs anyone in a bin that holds grain	Installation sequence review
R12	Electrical isolation	Controller, cable and pods at 15 V DC or less; the relay kit is the only mains-connected item, installed by a licensed electrician, with 2.5 kV or more isolation between control input and contacts	Circuit review; relay datasheet
R13	Affordable		

ID	Requirement	Target	Verification (TRL 3 or later)
		\$250 or less in parts per bin, with the farmhouse receiver (one per farm) costed per farm and excluded; priced options excluded	Priced BOM (bom/ bom.csv)
R14	Local data	All readings and fan hours stored on the farm; no cloud account needed	Architecture review

Table 2. Status at TRL 3 (GGD-CAL-001, Table 3). Not met items first.

ID	Status at TRL 3	Basis
R8	Not met	Autonomy 7.7 days with the fan continuous and a 3 mA relay input (met); refill from 50 % takes 7.3 days with the 10 W panel (not met); a panel of about 19 W would meet it (open, GGD-DDR-001 O2)
R2	At risk	Sensor-only error 0.23 points (SHT45) and 0.42 points (SHT40) worst case over 20 % to 75 % RH, 0.81 at the SHT40 maximum tolerance; equation fit error, drift and membrane lag not included
R4	At risk	Rule met by design; fan heat is 0.5 to 2.7 °C depending on the fan and moves the plenum EMC about 1 point per °C; the plenum probe option removes this
R6	At risk	Alarm latency about 13 min (met); only the center core, 3.3 % of the cross-section, is watched
R9	At risk	AGM electrolyte may freeze near -25 °C at 50 % charge; phosphine protection unverified
R3	Not verifiable at TRL 3	SHT45 meets the target; the radiation error of the shield needs a test
R10	Not verifiable at TRL 3	Rope 8 times the 2.5 kN design force (estimate 2.08 kN); the roof hanger rating comes from the bin maker
R1	Met	Six pods at 0.94 m pitch; ± 0.1 °C (SHT45) and ± 0.2 °C (SHT40) typical
R5	Met	Normally open relay, stale-data timeout and hand-off-auto switch
R7	Met	17.6 dB margin at 1 km past one farm building (plane-earth model); 24-byte packets fit the 400 ms dwell limit
R11	Met	Empty-bin installation sequence in GGD-PRC-001
R12	Met	12 V system with charging clamped at 15.0 V; optically isolated relay input
R13	Met	\$247.00 per bin; receiver \$22.00 per farm
R14	Met	Farmhouse receiver stores data locally

■ Assumptions

- Grain properties: shelled corn at 721 kg/m³ (25.4 kg per 0.0352 m³ bushel) and a specific heat of about 2.0 kJ/(kg·K) at 15 % moisture.
- EMC from the ASABE D245.6 modified Henderson equation for corn; accuracy of RH-based moisture prediction as reported by [USDA ARS](#).
- The fan warms the air by its input power over the air's heat capacity rate: about 1.0 °C for a 0.40 kW fan, 0.5 to 2.7 °C for 0.2 to 1.1 kW (GGD-CAL-001, section A).
- Cooling front time of 75 to 119 h at 0.2 cfm/bu, from extension guidance and a sensible-heat balance.

- Design pull-down force of 2.5 kN for a thin 6 mm cable is kept; GGD-CAL-001 estimates about 2.08 kN with an unloading overpressure factor of 2.0. Forces up to 4.7 kN were measured on the largest commercial cable during unloading ([Illinois Experts](#), [ASABE](#)); thinner cables see less force, but this is unverified.